

Sprint 4 Plan

Product Name: Spatial Layout Engine

Team Name: Vision Studio

Sprint Completion Date: Jun 5, 2026

Revision Number: 1.0

Revision Date: May 20, 2026

Goal: Sprint 4 focuses on giving users an immersive, shareable preview of their redesigned room. This sprint delivers the 3D visualization pipeline, export capabilities, and external integrations so users can experience their space in full 3D and bring their designs into professional tools or share them with others.

Tasks

1. As a user, I want to walk through my designed room in 3D so that I can understand how it will truly feel.
 - 1.1. Set up Three.js / React Three Fiber rendering environment inside the editor. (2.5 hours)
 - 1.2. Convert 2D room layout data (walls, dimensions) into extruded 3D geometry. (3 hours)
 - 1.3. Render floor, ceiling, and wall geometry from the JSON room state. (2 hours)
 - 1.4. Add basic first-person or orbit camera navigation for the 3D view. (2 hours)
 - 1.5. Test 3D rendering across at least two saved room layouts. (1.5 hours)

Total for 3D Room Visualization: 11 hours

2. As a user, I want realistic furniture shown in 3D so that my room feels complete.
 - 2.1. Map placed 2D furniture items to 3D furniture models or placeholder geometry. (2.5 hours)
 - 2.2. Load and render furniture assets (GLB/FBX) in the 3D scene at correct positions and rotations. (3 hours)
 - 2.3. Add material/color properties to furniture models to match catalog metadata. (2 hours)
 - 2.4. Test furniture rendering with at least five furniture types (sofa, bed, table, chair, desk). (1.5 hours)

Total for 3D Furniture Rendering: 9 hours

3. As a user, I want to move freely through the room so that I can explore every angle.
 - 3.1. Implement smooth camera controls (pan, orbit, zoom) for the 3D viewer. (2 hours)
 - 3.2. Add collision detection so the camera cannot pass through walls or furniture. (2.5 hours)

- 3.3. Add a top-down 2D minimap overlay for orientation within the 3D view. (2 hours)
- 3.4. Test navigation usability across different room sizes and furniture densities. (1.5 hours)

Total for 3D Camera Navigation: 8 hours

- 4. As a user, I want polished lighting and visuals so that the room feels exciting and realistic.
 - 4.1. Add ambient and directional lighting to the 3D scene. (1.5 hours)
 - 4.2. Add shadow casting for furniture and walls. (2 hours)
 - 4.3. Apply basic material shading (matte, wood, fabric) to room surfaces and furniture. (2 hours)
 - 4.4. Add a wall color/wallpaper overlay from the user's room customization settings into 3D. (1.5 hours)
 - 4.5. Test visual quality and performance across device sizes. (1.5 hours)

Total for Lighting and Visuals: 8.5 hours

- 5. As a user, I want to export my layout so that I can use or share it outside of Vision Studio.
 - 5.1. Build a JSON export function that captures full room and furniture state. (2 hours)
 - 5.2. Add a GLB/GLTF export pipeline for the 3D scene. (3 hours)
 - 5.3. Create an export UI panel with format options (JSON, GLB) and a download button. (1.5 hours)
 - 5.4. Test that exported files load correctly in external tools (e.g., Blender, Planner 5D). (2 hours)

Total for Export: 8.5 hours

Total Sprint Estimated Time: 45 hours

Team Roles

- Ashley Kim: Product Owner, Frontend Developer, UX/UI Lead, Deliverables Manager
- William Liu: Full Stack Developer, Coding Lead, 3D Rendering Lead
- Sriya Katreddi: Backend Developer, Export Pipeline Lead, Integration Support
- Ethan Cao: Scrum Master, Backend Developer, 3D Camera & Navigation Lead

Task Assignment

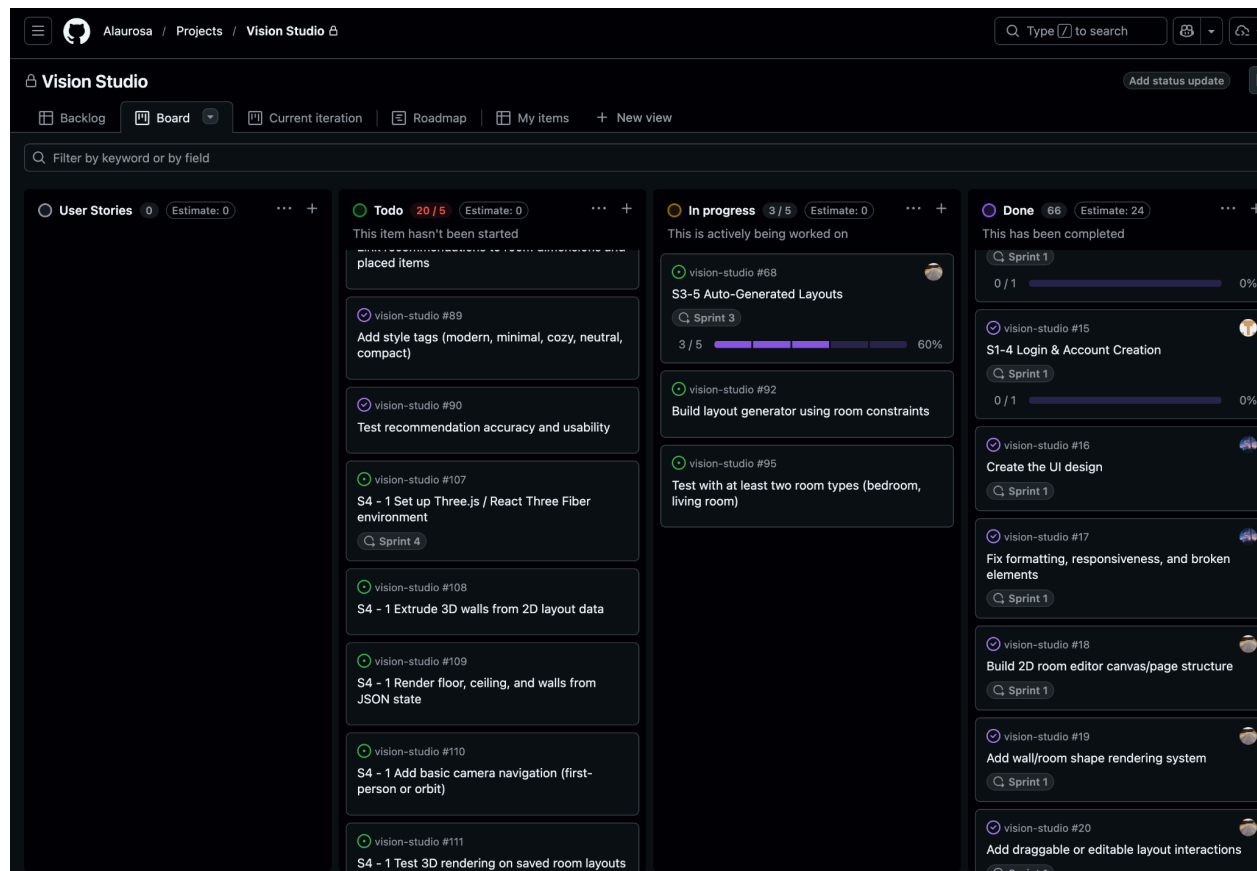
- Ashley Kim: User Story 1, Set up Three.js / React Three Fiber rendering environment
- William Liu: User Story 2, Map placed 2D furniture items to 3D models and load GLB assets
- Sriya Katreddi: User Story 5, Build JSON export function and GLB/GLTF export pipeline
- Ethan Cao: User Story 3, Implement smooth camera controls for the 3D viewer

Backlog / Flex Assignment: Whoever finishes first will begin User Story 4, Add ambient and directional lighting to the 3D scene.

Burnup Chart

[Burnup chart to be added here]

Scrum Board



Scrum Times

- Scrum Meeting 1: Monday 4:00 PM - 5:00 PM
- TA Meeting: Tuesday 11:00 AM - 12:00 PM
- Scrum Meeting 2: Tuesday 12:00 PM - 1:00 PM
- Scrum Meeting 3: Thursday 1:30 PM - 2:30 PM